1. **The online train reservation system allows users to search for trains by route, date, and time, and view available seats and classes. Users can register, log in, and manage their reservations through a personal dashboard. The booking process involves selecting seats, entering passenger details, and making secure payments via an integrated payment gateway. Admins can manage train schedules, seat availability, and user accounts through a comprehensive admin panel. Users receive booking confirmation emails and can view their booking history and print tickets from their dashboard. The website features real-time seat availability updates and is fully responsive for seamless use on all devices.**
   The Reservation system should contain the following features:

    1. If a passenger wants to reserve ticket(s), firstly, he/she has to log in to the Railway system with valid credentials. Then, the passenger has to provide his/her details with the date of the journey, names of the passengers and their details, origin station details, destination station details, and the class type of the required ticket(s).
    2. The Railway Reservation System will provide the available Train-list, and Seat- availability, via-details.
    3. To book a ticket passengers can pay through online/offline mode. After successful payment of the ticket fare the System will generate the ticket and PNR no. will be given to the passenger. The System also keeps the payment details and sends them to the system Admin.
    4. The Passenger can check PNR status (confirmed, RAC, waiting list) by entering the PNR no. into the Reservation system.
    5. The Reservation system should store all train details, fare details (by zone, class, and date wise), PNR no, date of trains, etc. This maintenance should be controlled by the Admin.
    6. The System also has refund rules which have a date of reservation, ticket fare, and refundable percentage. The passenger can simply cancel the ticket(s) by entering the PNR no and a cancel ticket request. After cancellation, the admin will pass the refundable amount to the System and the System will give the refundable amount to the passenger.

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Online Train Reservation System</title>
<style>
body {
   font-family: Arial;
   margin: 0;
   background: #f1f4f8;
}
```

```
header {
    background: #164477;
    color: white;
    text-align: center;
    padding: 22px;
}
nav {
    background: #102d50;
    padding: 12px;
    text-align: center;
}
nav button {
    padding: 10px 15px;
    margin: 5px;
    border: none;
    cursor: pointer;
    border-radius: 5px;
}
.container {
    width: 85%;
    margin: 25px auto;
    background: white;
    padding: 22px;
    border-radius: 10px;
}
input, select {
    width: 90%;
    padding: 10px;
    margin: 8px;
}
.action {
    background: #164477;
    color: white;
    padding: 10px 15px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}
.train {
    background: #e9f1fa;
    padding: 15px;
    margin: 12px;
    border-radius: 8px;
}
section {
    display: none;
```

```
}
#home {
   display: block;
}
</style>
</head>
<body>
<header>
<h1>Online Train Reservation System</h1>
<p>Search and Reserve Train Tickets Online</p>
</header>
<nav>
<button onclick="show('home')">Home</button>
<button onclick="show('login')">Login</button>
<button onclick="show('search')">Search Trains</button>
<button onclick="show('pnr')">PNR Status</button>
<button onclick="show('tickets')">My Tickets</button>
<button onclick="show('admin')">Admin</button>
</nav>
<div class="container">
<section id="home">
<h2>Welcome to Railway Reservation</h2>
<p>Book your train tickets easily and comfortably.</p>
<h3>Our Services</h3>
<ul>
<li>Train Search</li>
<li>Seat Availability</li>
<li>Online Ticket Booking</li>
<li>PNR Status Checking</li>
<li>Ticket Cancellation</li>
<li>Refund Calculation</li>
</ul>
</section>
<section id="login">
<h2>Passenger Login / Registration</h2>
<input type="text" id="username"
placeholder="Enter Passenger Name">
<input type="email" id="email"
placeholder="Enter Email">
<input type="password" id="password"
placeholder="Enter Password">
<button class="action" onclick="login()">
Login / Register
</button>
<p id="loginMsg"></p>
</section>
```

```
<section id="search">
<h2>Search Available Trains</h2>
<label>Origin Station</label>
<input type="text" id="from"
placeholder="Enter Source Station">
<label>Destination Station</label>
<input type="text" id="to"
placeholder="Enter Destination Station">
<label>Journey Date</label>
<input type="date" id="date">
<label>Travel Class</label>
<select id="classType">
<option>Sleeper</option>
<option>3A</option>
<option>2A</option>
<option>1A</option>
</select>
<button class="action" onclick="searchTrain()">
Search Trains
</button>
<div id="trainList"></div>
</section>
<section id="pnr">
<h2>Check PNR Status</h2>
<input type="text" id="pnrNumber"
placeholder="Enter PNR Number">
<button class="action" onclick="checkPNR()">
Check Status
</button>
<p id="pnrResult"></p>
</section>
<section id="tickets">
<h2>My Reservation History</h2>
<div id="ticketList"></div>
</section>
<section id="admin">
<h2>Admin Dashboard</h2>
<h3>Add New Train</h3>
<input type="text" id="trainName"
placeholder="Train Name">
<input type="text" id="trainFrom"
placeholder="Source Station">
<input type="text" id="trainTo"
placeholder="Destination Station">
```

```
<input type="number" id="trainFare"
placeholder="Ticket Fare">
<input type="number" id="trainSeats"
placeholder="Available Seats">
<button class="action" onclick="addTrain()">
Add Train
</button>
<h3>Manage Trains</h3>
<div id="adminList"></div>
<h3>Booking Report</h3>
<button class="action" onclick="report()">
Generate Report
</button>
<p id="reportResult"></p>
</section>
</div>
<script>
let trains = [
{
name:"Chennai Express",
from:"Chennai",
to:"Mumbai",
fare:1200,
seats:50
},
{
name:"Vaigai Express",
from:"Chennai",
to:"Madurai",
fare:450,
seats:40
},
{
name:"Pandian Express",
from:"Chennai",
to:"Madurai",
fare:650,
seats:35
}
];
let tickets = [];
let currentUser = "";
function show(id) {
document.querySelectorAll("section").forEach(s=>{
s.style.display="none";
});
```

```
document.getElementById(id).style.display="block";
if(id==="admin") adminTrains();
if(id==="tickets") viewTickets();
}
function login() {
let name=document.getElementById("username").value;
let password=document.getElementById("password").value;
if(name.trim()==="" || password.trim()==="") {
alert("Please enter login details");
return;
}
currentUser=name;
document.getElementById("loginMsg").innerHTML=
"Welcome "+name+"! Login Successful.";
}
function searchTrain() {
let from=document.getElementById("from").value.toLowerCase();
let to=document.getElementById("to").value.toLowerCase();
let date=document.getElementById("date").value;
if(!from || !to || !date) {
alert("Enter all details");
return;
}
let result=trains.filter(t=>
t.from.toLowerCase()===from &&
t.to.toLowerCase()===to &&
t.seats>0
);
let output="";
result.forEach(t=>{
let index=trains.indexOf(t);
output+=`
<div class="train">
<h3>${t.name}</h3>
<p>From: ${t.from}</p>
<p>To: ${t.to}</p>
<p>Journey Date: ${date}</p>
<p>Available Seats: ${t.seats}</p>
<p>Fare: ₹${t.fare}</p>
<button class="action"
onclick="bookTicket(${index})">
Book Ticket
</button>
</div>`;
```

```
});

document.getElementById("trainList").innerHTML=
output || "No trains available.";
}
function bookTicket(index) {
if(!currentUser) {
alert("Please login first");
show("login");
return;
}
let train=trains[index];
let passenger=prompt("Enter Passenger Name:");
if(!passenger || !passenger.trim()) return;
let count=Number(prompt("Enter Number of Tickets:"));
if(!Number.isInteger(count) || count<=0 ||
count>train.seats) {
alert("Invalid seat availability");
return;
}
let date=document.getElementById("date").value;
let travelClass=document.getElementById("classType").value;
let pnr=Math.floor(1000000000+Math.random()*9000000000);
let ticket={
pnr:pnr,
user:currentUser,
passenger:passenger,
train:train.name,
from:train.from,
to:train.to,
date:date,
classType:travelClass,
seats:count,
fare:train.fare*count,
status:"Confirmed"
};
train.seats-=count;
tickets.push(ticket);
alert(
"Ticket Booked Successfully!\n"+
"PNR Number: "+pnr+
"\nTotal Fare: ₹"+ticket.fare
);
searchTrain();

}
```

```
function checkPNR() {
let number=document.getElementById("pnrNumber").value;
let ticket=tickets.find(t=>t.pnr==number);
if(ticket) {
document.getElementById("pnrResult").innerHTML=
"PNR: "+ticket.pnr+
"<br>Passenger: "+ticket.passenger+
"<br>Train: "+ticket.train+
"<br>Journey Date: "+ticket.date+
"<br>Class: "+ticket.classType+
"<br>Status: "+ticket.status;

} else {

document.getElementById("pnrResult").innerHTML=
"Invalid PNR Number.";
}
}
function viewTickets() {
let output="";
tickets.forEach((t,i)=>{
if(t.user===currentUser) {
output+=`
<div class="train">
<h3>${t.train}</h3>
<p>PNR Number: ${t.pnr}</p>
<p>Passenger: ${t.passenger}</p>
<p>Route: ${t.from} to ${t.to}</p>
<p>Date: ${t.date}</p>
<p>Class: ${t.classType}</p>
<p>Tickets: ${t.seats}</p>
<p>Fare: ₹${t.fare}</p>
<p>Status: ${t.status}</p>

${t.status==="Confirmed"?
`<button class="action"
onclick="cancelTicket(${i})">
Cancel Ticket
</button>`:""}
</div>`;
}
});

document.getElementById("ticketList").innerHTML=
output || "No tickets booked.";
```

```
}
function cancelTicket(index) {
if(confirm("Do you want to cancel this ticket?")) {
let ticket=tickets[index];
let train=trains.find(t=>t.name===ticket.train);
if(train) train.seats+=ticket.seats;
let refund=ticket.fare*0.80;
ticket.status="Cancelled";
alert("Ticket Cancelled Successfully!\nRefund: ₹"+refund);
viewTickets()
}
}
function addTrain() {
let name=document.getElementById("trainName").value;
let from=document.getElementById("trainFrom").value;
let to=document.getElementById("trainTo").value;
let fare=Number(document.getElementById("trainFare").value);
let seats=Number(document.getElementById("trainSeats").value);

if(!name || !from || !to || fare<=0 ||
!Number.isInteger(seats) || seats<=0) {
alert("Enter valid train details");
return;
}
trains.push({
name:name,
from:from,
to:to,
fare:fare,
seats:seats
});
alert("Train Added Successfully");
adminTrains();
}
function adminTrains() {
document.getElementById("adminList").innerHTML=
trains.map((t,i)=>`
<div class="train">
<b>${t.name}</b>
<p>${t.from} to ${t.to}</p>
<p>Fare: ₹${t.fare}</p>
<p>Available Seats: ${t.seats}</p>
<button class="action"
onclick="deleteTrain(${i})">
Delete Train
```

```
</button>
</div>
`).join("");

}

function deleteTrain(index) {

if(confirm("Delete this train?")) {
trains.splice(index,1);
adminTrains();
}

}

function report() {
document.getElementById("reportResult").innerHTML=
"Total Trains: "+trains.length+
"<br>Total Bookings: "+tickets.length+
"<br>Total Revenue: ₹"+
tickets.reduce((sum,t)=>sum+t.fare,0);

}
</script>
</body>
</html>
```

Output:

